

considers the category monthly change (CMC), which represents the monthly percentage change in the effectiveness of the feature's categories. To calculate MFEM, the CMC is multiplied by the respective category weightage (CWT) and then divided by the sum of CWTs for all four categories — DAU, SD, CR, and CLV. This calculation results in a single numerical value that indicates the feature's overall effectiveness for that month. For instance, if your feature's DAU, SD, CR, and CLV categories have distinct changes in a given month, MFEM provides an aggregated measure that considers these changes in relation to their individual importance (CWT) for the feature.

$$MFEM = \frac{(CMC * CWT)}{(DAU\ CWT + SD\ CWT + CR\ CWT + CLV\ CWT)}$$

- **Quarterly FEM (QFEM):** The quarterly FEM is a broader evaluation metric that is calculated at the end of every quarter. It offers a more extensive perspective on the feature's effectiveness by considering the monthly FEM values over the three months of a quarter (M1 MFEM, M2 MFEM, and M3 MFEM). The QFEM is derived by summing these monthly FEM values and then dividing the result by three to calculate the quarterly average. For example, if you want to assess the feature's performance over an entire quarter, QFEM provides a summary value based on the changes observed in the monthly FEM values. This quarterly measure is especially useful for longer-term performance evaluation.

$$QFEM = \frac{M1\ MFEM + M2\ MFEM + M3\ MFEM}{3}$$

Feature standard deviation (FSD)

This statistical measure quantifies the level of variation or fluctuation in the FEM across all the data points. It serves as an important metric to assess how consistent or stable the feature's performance has been over a specific period. A small standard deviation indicates that the effectiveness of the feature remains relatively stable and exhibits less variation over the determined range, which is often considered desirable. In contrast, a large standard deviation suggests that the effectiveness of the feature is highly volatile, which may not be preferable as it can lead to inconsistent outcomes. The calculation of FSD can be conducted over various timeframes, with a common practice being to calculate it over all the data points collected during a period of one year. However, this calculation can be extended to larger datasets for more comprehensive insights.

- **Monthly standard deviation (MSD):** This denotes a variation of the feature standard deviation specifically calculated at the end of every month. It quantifies the amount of variation or fluctuation in the monthly feature effectiveness measure (MFEM) values over a series of months. MSD offers insights into how consistent or variable the feature's effectiveness has been monthly. To calculate MSD, the formula involves finding the square root of the sum of squared differences between each monthly MFEM data point (x) and the average of the monthly MFEM values (μ). This value is then divided by the total number of data points (n) to obtain the monthly standard deviation.

$$MSD = \sqrt{\frac{\sum_{i=1}^n (x_i - \mu)^2}{n}}$$

- μ = Average of MFEM
- x = Each monthly data point
- n = Total data points

- **Quarterly standard deviation (QSD):** This is designed to measure the variation in the FEM over a broader timeframe, specifically at the end of every quarter. It provides a more comprehensive perspective on the feature's stability or variability by considering the monthly FEM values over the three months of a quarter. The formula for calculating QSD is akin to MSD, involving the square root of the sum of squared differences between each quarterly FEM data point (x) and the average of the quarterly FEM values (μ). The result is divided by the total number of data points (n) to obtain the quarterly standard deviation.

$$QSD = \sqrt{\frac{\sum_{i=1}^n (x_i - \mu)^2}{n}}$$

- μ = Average of QFEM
- x = Each quarterly data point
- n = Total data points

Feature absolute deviation (FAD)

This statistical measure quantifies the extent to which the FEM values deviate from the mean. It provides insights into how much the effectiveness of a feature varies from the average value. A small absolute deviation indicates that the effectiveness of the feature remains relatively close to the mean over the determined range, which is often considered desirable. Conversely, a large